

Akhil N

Full Stack Developer – MERN

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Summary

Full Stack Developer with hands-on experience building production-grade MERN stack applications using microservices architecture, Docker, Kubernetes, and AWS EKS. Proficient in React.js, Node.js, Express.js, MongoDB, and PostgreSQL. Experienced in designing RESTful APIs, implementing real-time features with WebSocket and WebRTC, integrating payment gateways, and automating CI/CD pipelines with GitHub Actions. Consistently applies clean architecture and SOLID principles across all projects.

Technical Skills

- **Programming Languages:** JavaScript (ES6+), TypeScript, C
- **Frontend:** React.js, Redux, Tailwind CSS, ShadCN UI, HTML5, CSS3, EJS
- **Backend:** Node.js, Express.js, REST API, TypeORM, gRPC, JWT Authentication, WebSocket
- **Databases:** MongoDB, PostgreSQL, Redis
- **Message Queues & Real-Time:** Apache Kafka, Socket.IO, WebRTC
- **Cloud & DevOps:** AWS (EC2, S3, EKS), Docker, Kubernetes, Helm, Skaffold, NGINX
- **CI/CD & Version Control:** Git, GitHub Actions
- **Testing & Monitoring:** Jest, Sentry, ELK Stack (Elasticsearch, Logstash, Kibana)
- **Architecture:** Microservices, Clean Architecture, MVC, SOLID Principles, Event-Driven Architecture

Projects

Buxlo – Full Stack Money Management & Mentorship Platform

[Live Link](#) | [GitHub](#)

A microservices-based web application for personal finance management and expert mentorship, featuring real-time video calls, session scheduling, and secure payments.

- Developed a responsive frontend using **React.js**, **TypeScript**, **Tailwind CSS**, and **ShadCN UI**, with interactive financial flow visualizations via **React Flow**.
- Integrated real-time **peer-to-peer video calling and chat** using **WebRTC** data channels for mentor-user communication.
- Architected a **microservices backend** with **Node.js** and **Express.js**, covering user management, session scheduling, and content delivery.
- Designed a custom **API Gateway** for centralized routing, rate limiting, and structured logging across all services.
- Configured **NGINX Ingress Controller** in **Kubernetes** for service discovery, load balancing, and high availability.
- Established **event-driven inter-service communication** using **Apache Kafka** and synchronous calls via **gRPC**.
- Engineered a session booking system with **Redis** and **Bull.js** for distributed job queuing and background task processing.
- Integrated **Stripe** payment processing with webhook handling and transaction validation.
- Optimized data operations using **PostgreSQL** with **TypeORM**, **MongoDB** for unstructured data, and **Redis** for caching.
- Set up **Sentry** for real-time error tracking and performance monitoring in production.
- Automated **CI/CD pipelines** with **GitHub Actions** for continuous testing, image builds, and Kubernetes

deployments.

- **Technologies:** React.js, TypeScript, Node.js, Express.js, MongoDB, PostgreSQL, Redis, Kafka, WebRTC, Docker, Kubernetes, AWS EKS, Sentry, Stripe

ECOM-Bicycle – Full Stack E-Commerce Platform

[Live Link](#) | [GitHub](#)

An online bicycle store where users can browse products, manage carts, and complete secure purchases.

- Developed a responsive frontend using **EJS** templating with custom **CSS** and **Bootstrap**.
- Created an advanced product catalog with filtering, search, and dynamic cart management.
- Implemented user authentication with **JWT**, session management, and **bcrypt** password hashing.
- Designed **RESTful APIs** with **Node.js** and **Express.js** for order processing and inventory management.
- Modeled and optimized a **MongoDB** schema for efficient querying of products, orders, and user data.
- Integrated **Razorpay** payment gateway with webhook-based signature verification and transaction validation.
- Deployed on **AWS EC2** with **NGINX** as a reverse proxy for SSL termination and static asset serving.
- **Technologies:** Node.js, Express.js, MongoDB, EJS, Bootstrap, AWS EC2, NGINX, Razorpay API

Additional Projects

- **RepoRadar (GitHub Repository Finder):** Developed a **React.js** SPA that lets users search GitHub repositories by username, showing repository details and direct links via a clean UI. [Live Link](#) | [GitHub](#)
- **Ticketing Microservice System:** Designed a full stack ticketing platform with **microservices architecture**, **event-driven** communication via **Kafka**, and **Stripe** payment processing. [GitHub](#)
- **To-Do List Application:** Created a task management app using **HTML**, **CSS**, and **JavaScript** with a clean interface for organizing daily tasks. [GitHub](#)

Education

Full Stack Web Development Brototype, Calicut	2025 – 2026
Electronic Mechanical (ITI) ITI Kozhikode	2022 – 2024
Higher Secondary – Humanities G.G.V.H.S Kallay	2020 – 2022